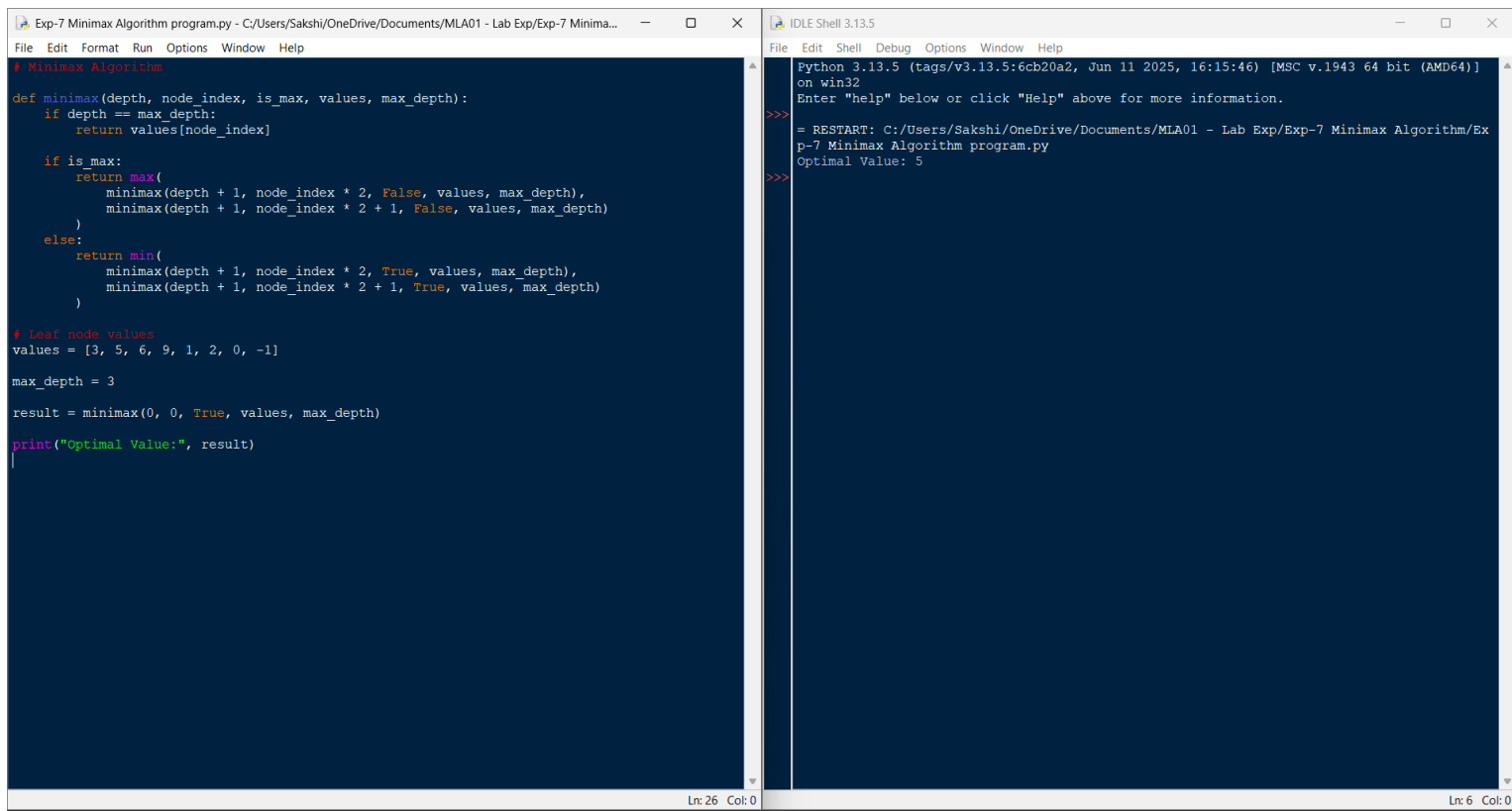


Exp:7 Minimax Algorithm

Program:

```
def minimax(depth, node_index, is_max, values, max_depth):  
    if depth == max_depth:  
        return values[node_index]  
    if is_max:  
        return max(  
            minimax(depth + 1, node_index * 2, False, values, max_depth),  
            minimax(depth + 1, node_index * 2 + 1, False, values, max_depth)  
        )  
    else:  
        return min(  
            minimax(depth + 1, node_index * 2, True, values, max_depth),  
            minimax(depth + 1, node_index * 2 + 1, True, values, max_depth)  
        )  
values = [3, 5, 6, 9, 1, 2, 0, -1]  
max_depth = 3  
result = minimax(0, 0, True, values, max_depth)  
print("Optimal Value:", result)
```

Output:



The image shows a screenshot of an IDE with two panes. The left pane displays a Python script for a Minimax algorithm, and the right pane shows the output of the program.

```
Exp-7 Minimax Algorithm program.py - C:/Users/Sakshi/OneDrive/Documents/MLA01 - Lab Exp/Exp-7 Minima...
File Edit Format Run Options Window Help

# Minimax Algorithm

def minimax(depth, node_index, is_max, values, max_depth):
    if depth == max_depth:
        return values[node_index]

    if is_max:
        return max(
            minimax(depth + 1, node_index * 2, False, values, max_depth),
            minimax(depth + 1, node_index * 2 + 1, False, values, max_depth)
        )
    else:
        return min(
            minimax(depth + 1, node_index * 2, True, values, max_depth),
            minimax(depth + 1, node_index * 2 + 1, True, values, max_depth)
        )

# Leaf node values
values = [3, 5, 6, 9, 1, 2, 0, -1]

max_depth = 3

result = minimax(0, 0, True, values, max_depth)

print("Optimal Value:", result)
|
```

Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

```
>>>
= RESTART: C:/Users/Sakshi/OneDrive/Documents/MLA01 - Lab Exp/Exp-7 Minimax Algorithm/Ex
p-7 Minimax Algorithm program.py
Optimal Value: 5
>>>
```

Ln: 26 Col: 0

Ln: 6 Col: 0